

Andrea Bonacci

AI ENGINEER — LLM & AGENTIC SYSTEMS

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PROFILE

From Data Scientist to AI Engineer, with over 4 years of experience in data-driven consulting for enterprise clients. I design and ship production AI systems: agents and agentic workflows with MCP and tool use, LLM integration (OpenAI, Anthropic), RAG and real-time voice AI. Solid background in end-to-end data analysis: pipelines, BI dashboards and NLP models, from ETL to deployment with FastAPI and Docker. Academic tutor for university interns and responsible for technical interviews for post-acquisition hiring. I also delivered an intensive 8-hour university course on AI applied to coding.

PROFESSIONAL EXPERIENCE

LINKALAB S.R.L. · CAGLIARI — part of Lynx Group since 2025 Nov 2021 → present
Consulting company — Internal growth

- AI Engineer & Data Analyst** Oct 2025 → present
AI Engineering focus: AI agents and agentic workflows with MCP, tool use, sub-agents and LLM integration (OpenAI, Anthropic). Production AI systems on client projects, including real-time conversational voice AI, agents for document and accounting automation, and RAG for compliance validation against regulatory corpora. Claude Code, Windsurf and Gemini CLI in production. Academic tutor for university interns. Technical interviews for hiring Data Scientists and AI Engineers post-acquisition.
AI Agents MCP LLM Integration Realtime Voice AI RAG
Document AI Claude Code Academic Tutor Technical Hiring
- Data Analyst & Data Scientist** Jul 2022 → Oct 2025
End-to-end data pipelines and BI dashboards (Power BI, QuickSight, Tableau, Looker Studio) for enterprise clients. NLP, sentiment analysis, time series. Deployment with FastAPI and Docker on AWS.
Python SQL Power BI QuickSight Tableau NLP FastAPI Docker
AWS S3/Athena
- Data Scientist** — contract Apr 2022 → Jun 2022
Post-internship hire. First autonomous data analysis and modeling projects for enterprise clients.
- Data Scientist** — curricular internship Nov 2021 → Mar 2022
Master's thesis internship. Development of an Italian Emotion Analysis system for online reputation monitoring.

AT A GLANCE

4+

years of professional experience

18+

open source projects

DIFFERENTIATORS

AI + BUSINESS

Business degree + Data Science master's = I build AI that solves real problems, from idea to deployment.

AI ENGINEERING

Advanced use of LLMs, AI agents and MCP to build intelligent workflows and automate complex tasks.

UNIVERSITY SPEAKER

AI applied to coding course, University of Cagliari, April 2026 — 8 hours.

MENTORING & HIRING

Academic tutor for university interns. Leading technical interviews for post-acquisition team expansion.

SKILLS

AI & LLM

AI Agents MCP Prompt Eng.

RAG OpenAI API Anthropic API

Tool Use Function Calling

Document AI Human-in-the-Loop

Realtime Voice Sub-agents

Hugging Face Ollama LLM Eval

Agentic Coding Config

DATA SCIENCE

Python SQL NLP

Sentiment Analysis Pandas

Time Series Computer Vision

BUSINESS INTELLIGENCE

Power BI QuickSight Tableau

Looker Studio

Design and delivery of two intensive sessions (8 hours total) on *"Artificial Intelligence Applied to Coding"*. LLM architecture, advanced prompt engineering, agentic coding with CLI tools, Model Context Protocol (MCP) integration and custom Skills creation for AI agents. Hands-on workshops where students built their first AI-assisted development workflows.

AI for Coding

Prompt Engineering

LLM

MCP

University Lecture

CLIENT PROJECTS

ON A CONTRACT BASIS FOR AN ENTERPRISE CLIENT · DETAILS ANONYMIZED FOR CONFIDENTIALITY

Real-Time Conversational Voice AI Systems

Low-latency, voice-to-voice AI conversation systems with automatic evaluation of the interactions.

- ▶ Integration of leading real-time voice AI providers (OpenAI Realtime API, Gemini Live) and streaming transcription (AssemblyAI).
- ▶ Multi-provider architectures with fallback and circuit breaker for production resilience.
- ▶ RAG knowledge bases (embeddings, vector search) queried by the AI via function calling.
- ▶ Automatic evaluation with anti-hallucination validation and multilingual prompt engineering, integrable into heterogeneous production stacks.

Realtime Voice AI

OpenAI Realtime API

RAG

Function Calling

Circuit Breaker

pgvector

AI Agents for Document & Finance Automation

AI agents for document and accounting process automation, such as automatic reconciliation of related documents.

- ▶ LLM agents with native function calling and validated structured output (Pydantic), with advanced prompt engineering.
- ▶ Document AI to extract structured data from heterogeneous documents (XML, PDF, digitally signed files) via parsing and information extraction.
- ▶ Deterministic matching engines with rules, tolerances and confidence scoring.
- ▶ Integration with business and ERP systems, plus human-in-the-loop flows for reviewing uncertain cases.

AI Agent

Function Calling

Document AI

Information Extraction

Fuzzy Matching

ERP Integration

RAG & Document Intelligence for Compliance

AI systems that validate document compliance (contracts, policies, playbooks) against a regulatory corpus, with source citation and human review.

- ▶ RAG pipelines with hybrid search (BM25 and semantic embeddings) and domain-specific chunking.
- ▶ Multi-phase agentic orchestration over LLMs for issue generation and structured verdicts with regulatory references.
- ▶ Reliability through validated output and anti-hallucination via mandatory back-links to sources.
- ▶ Human-in-the-loop flows with checkpoint, resume and audit trail; asynchronous FastAPI backend with WebSocket updates.

RAG

Hybrid Search

LLM Orchestration

Structured Output

Human-in-the-Loop

FastAPI

MLOPS & CLOUD

Docker

FastAPI

AWS S3

Athena

AWS Bedrock

pgvector

Flask

GitHub

LANGUAGES

Italian

Native

English

Fluent

AI & Automation

Ongoing

Two GitHub Actions on Marketplace with multi-provider LLM architecture and fallback: [AI PR Reviewer](#) (code review, key rotation) and [AI Changelog Generator](#) (release notes, self-evaluation, multilingual). [Video anonymization](#) with YOLO/CV, [Chrome extension for LLM-powered article/PDF summaries](#), [multilingual TTS](#).

- Python
- GitHub Actions
- GitHub Marketplace
- YOLO
- Computer Vision
- LLM API
- Chrome Extension

AI-Powered Web Apps

Ongoing

[BetTracker](#) — sports betting tracker with screenshot OCR, automatic result resolution, AI chat and advanced statistics. PWA with offline, OAuth and cloud sync.

- React
- TypeScript
- Gemini AI
- Firebase
- PWA
- Tailwind CSS

Full-Stack Web Apps

Ongoing

PWAs with cloud sync and offline: [finance](#), [health](#), [shared expenses](#). End-to-end ownership.

- JavaScript
- PWA
- Cloud Sync
- Responsive UI

Developer Tools & CLI

Ongoing

[Claude Code Tools Explorer](#) — TUI for config inspection (Textual/Rich). [Command Quiver](#) — desktop prompt & shell manager (GTK4, SQLite). [Telegram bot for remote multi-PC shell](#) (zero infrastructure, heartbeat, audit log, systemd), [file manager extensions](#), [clipboard image CLI](#), [audio renaming via AcoustID](#), and [smart clock-in notifications](#).

- Python
- JavaScript
- asyncio
- Textual/Rich
- Telegram Bot
- Shell
- Linux
- systemd
- CLI
- Chrome Extension

— EDUCATION

Master’s Degree — Data Science & Business Informatics

2022

University of Pisa
Thesis: Italian Emotion Analysis for online reputation monitoring

Bachelor’s Degree — Business Administration

2018

University of Pisa